Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.3208318268347284, median 0.25716064329256627, std: 0.24262550606068756
Gyroscope error (imu0): mean 0.17948290197305677, median 0.14599377259892782, std: 0.1434759289114345
Accelerometer error (imu0): mean 0.14881710814349638, median 0.11392767728698339, std: 0.1455248690501867

Residuals

Reprojection error (cam0) [px]: mean 0.3208318268347284, median 0.25716064329256627, std: 0.24262550606068756
Gyroscope error (imu0) [rad/s]: mean 0.0027886328937259384, median 0.0022683109759927458, std: 0.002229191139779834
Accelerometer error (imu0) [m/s^2]: mean 0.04665225226037326, median 0.035714863744736876, std: 0.04562017758429202

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99895635 -0.04552299 -0.00372329 0.00086992]
 [0.00332201 0.15371476 -0.98810968 -0.00060354]
 [0.04555404 0.98706607 0.15370557 -0.03263167]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[0.99895635 0.00332201 0.04555404 0.00061949]
 [-0.04552299 0.15371476 0.98706607 0.03234199]
 [-0.00372329 -0.98810968 0.15370557 0.00442254]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.009705672027625137

Gravity vector in target coords: [m/s^2]
[0.01334382 -9.80471339 -0.18931514]

Calibration configuration
==========

cam0

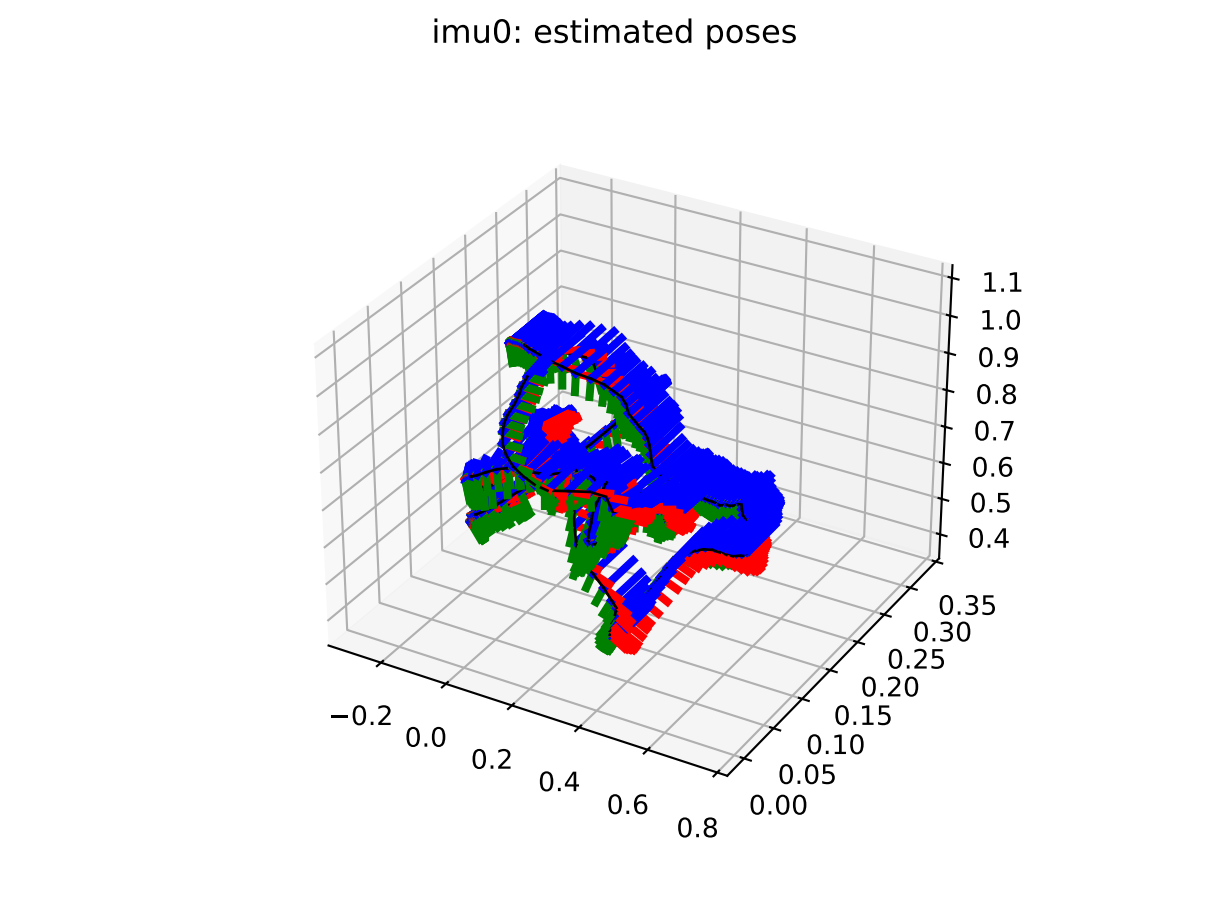
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.02 [m]
    Spacing 0.006 [m]

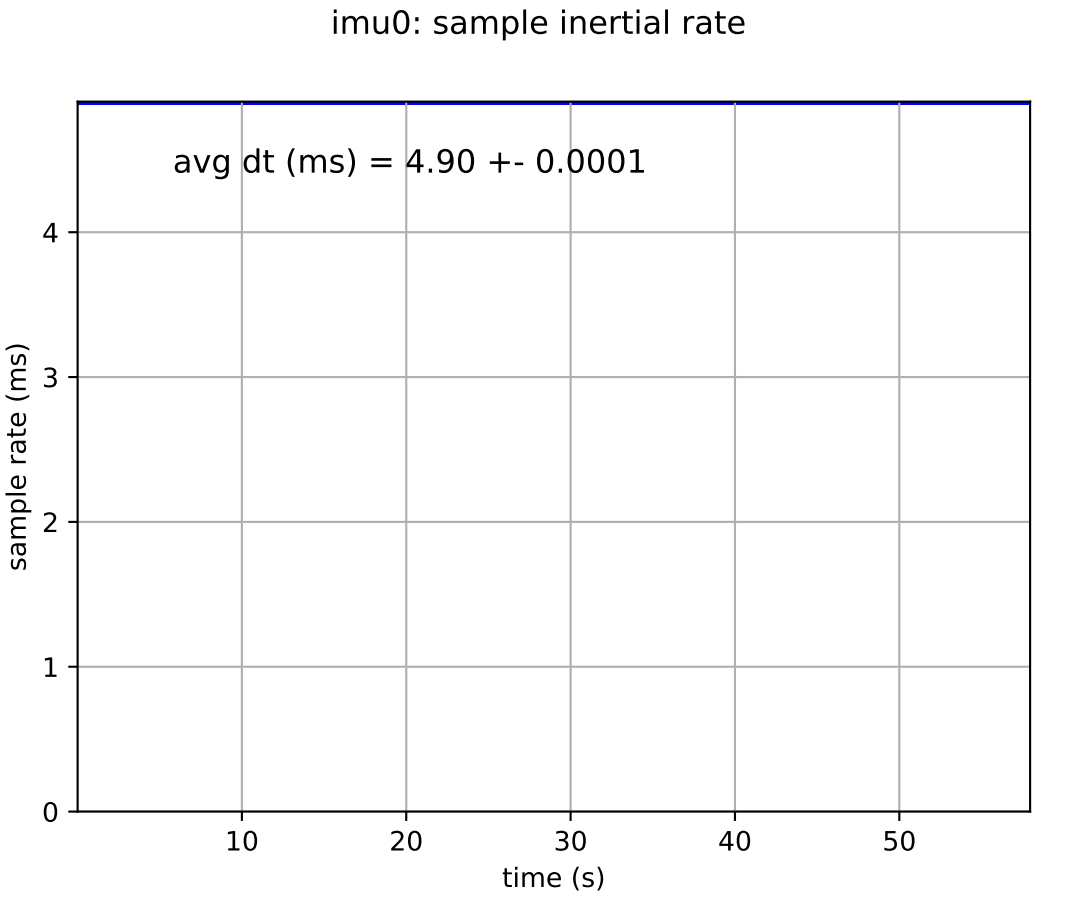
IMU configuration
==========

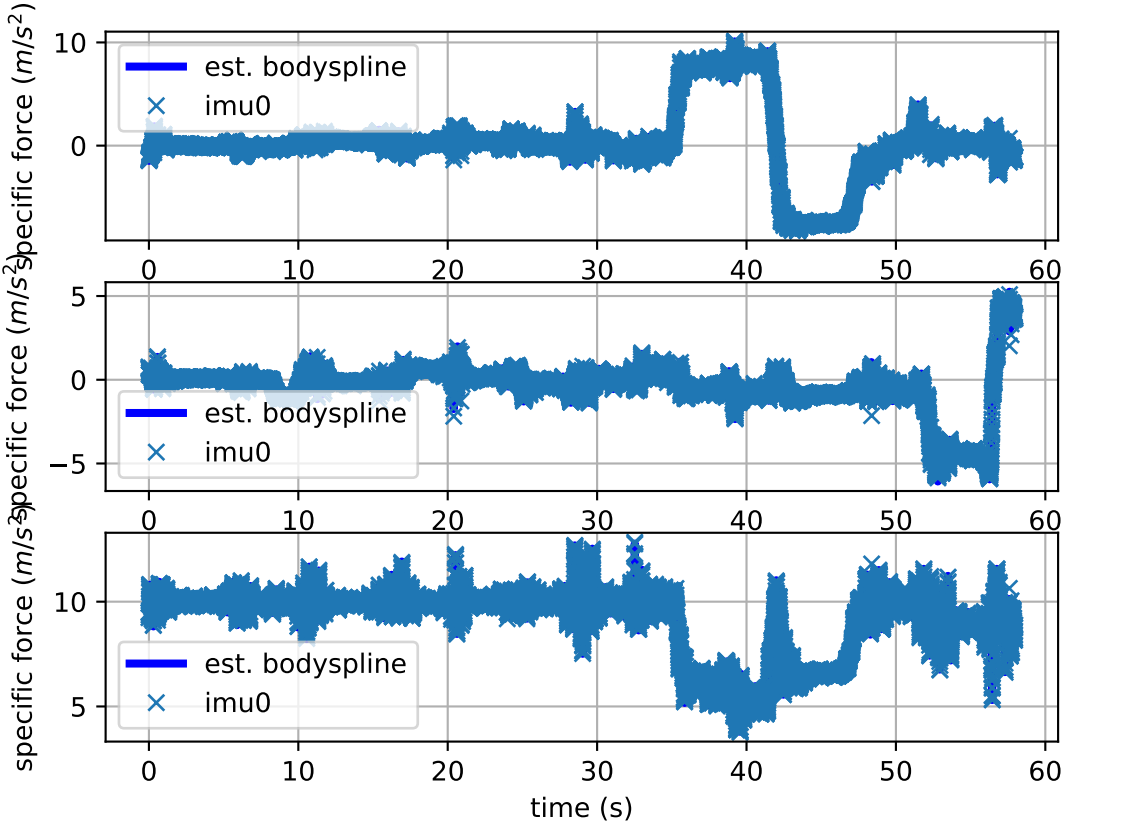
IMU0:

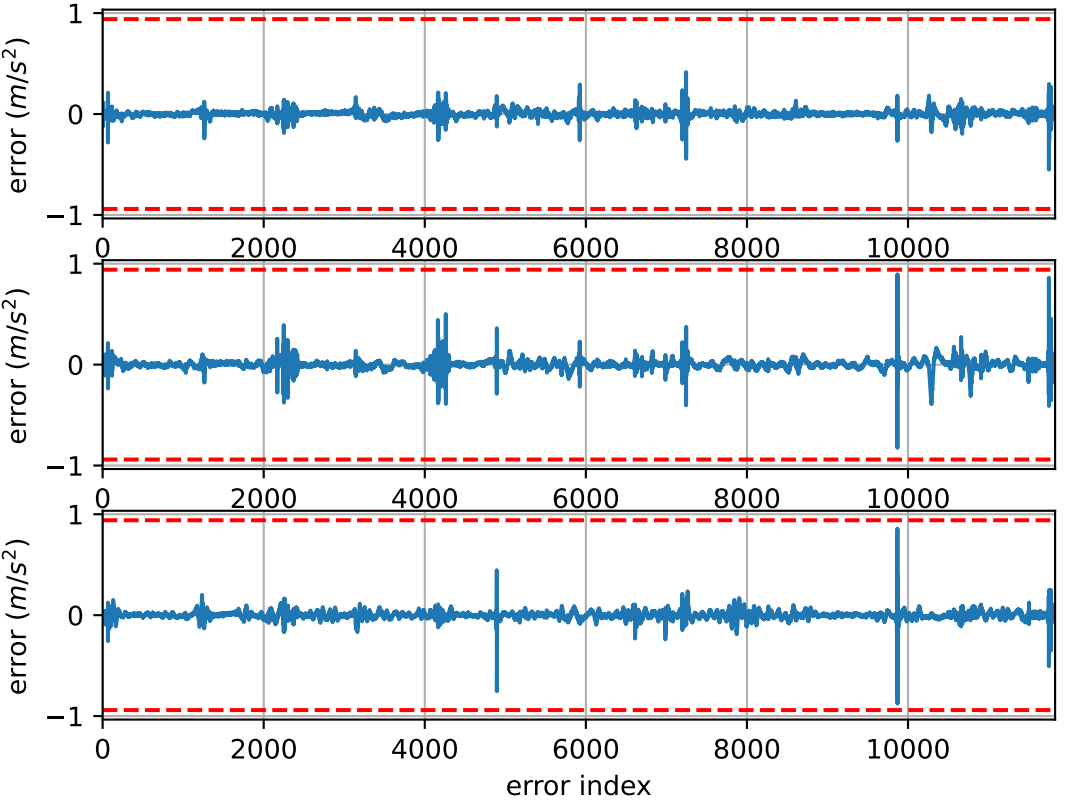
  Model: calibrated
  Update rate: 204.0355927763645
  Accelerometer:
    Noise density: 0.021946576707293677
    Noise density (discrete): 0.3134871577761677
    Random walk: 0.013079819165366822
  Gyroscope:
    Noise density: 0.0010877154466348196
    Noise density (discrete): 0.01553703925594292
    Random walk: 0.00011172744905164805

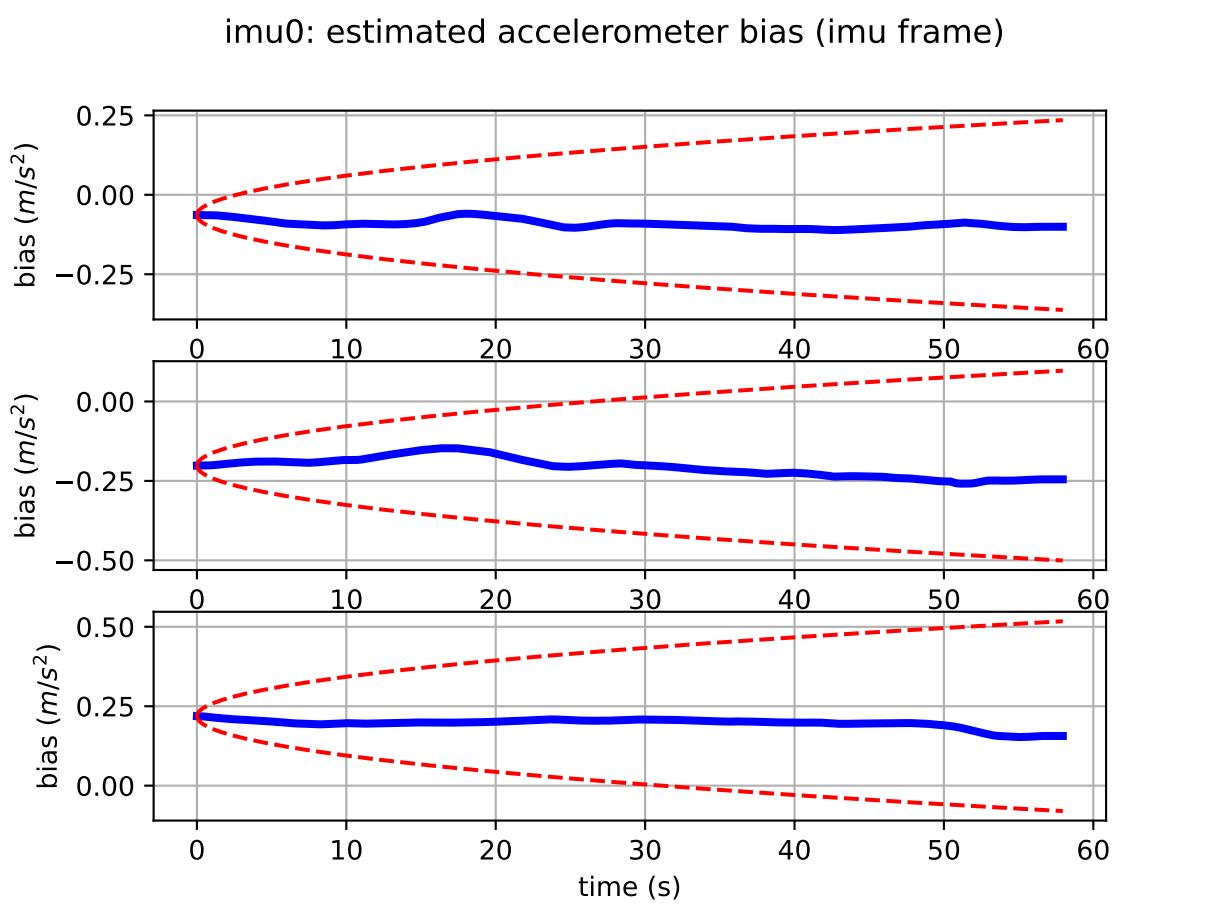
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

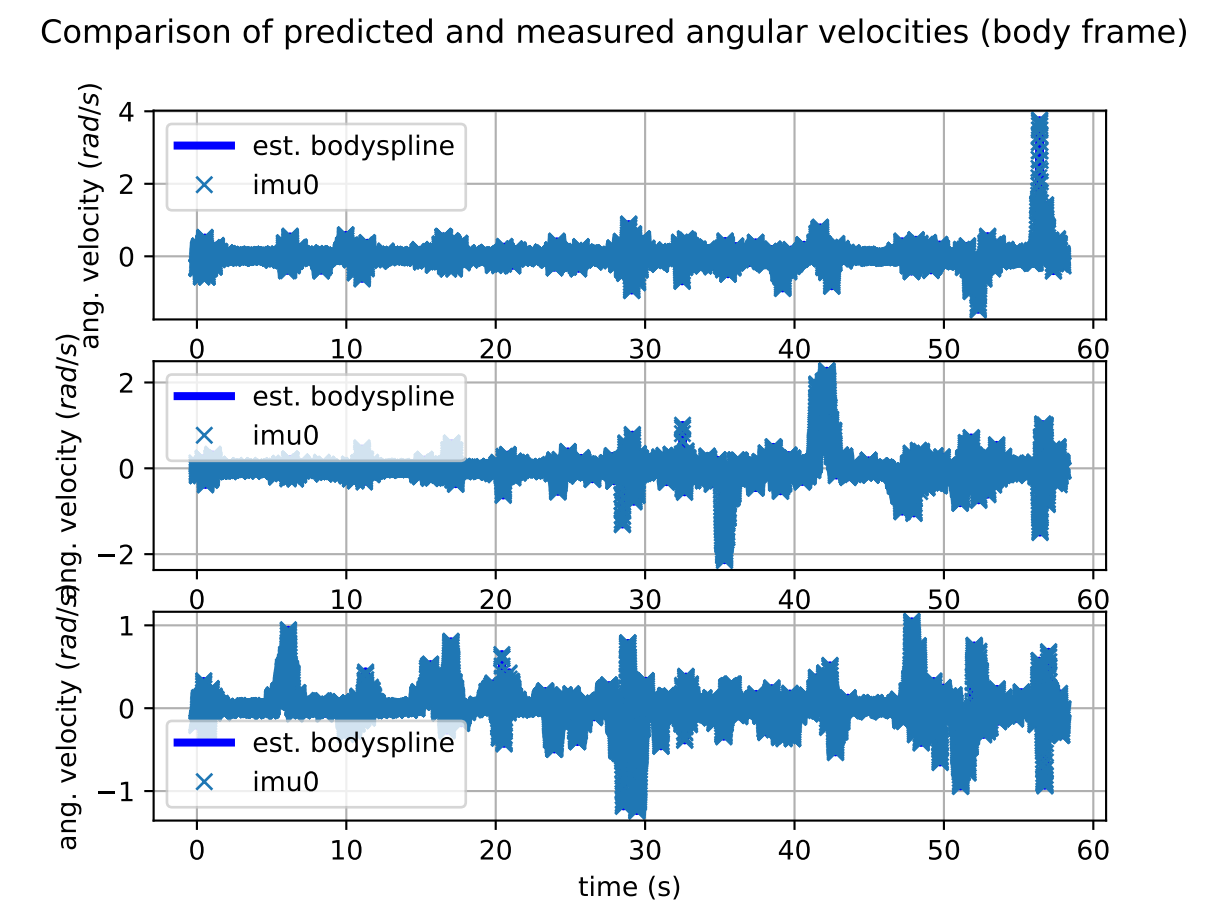

imu0: estimated poses
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.2
0.0
0.2
0.4
0.6
0.8

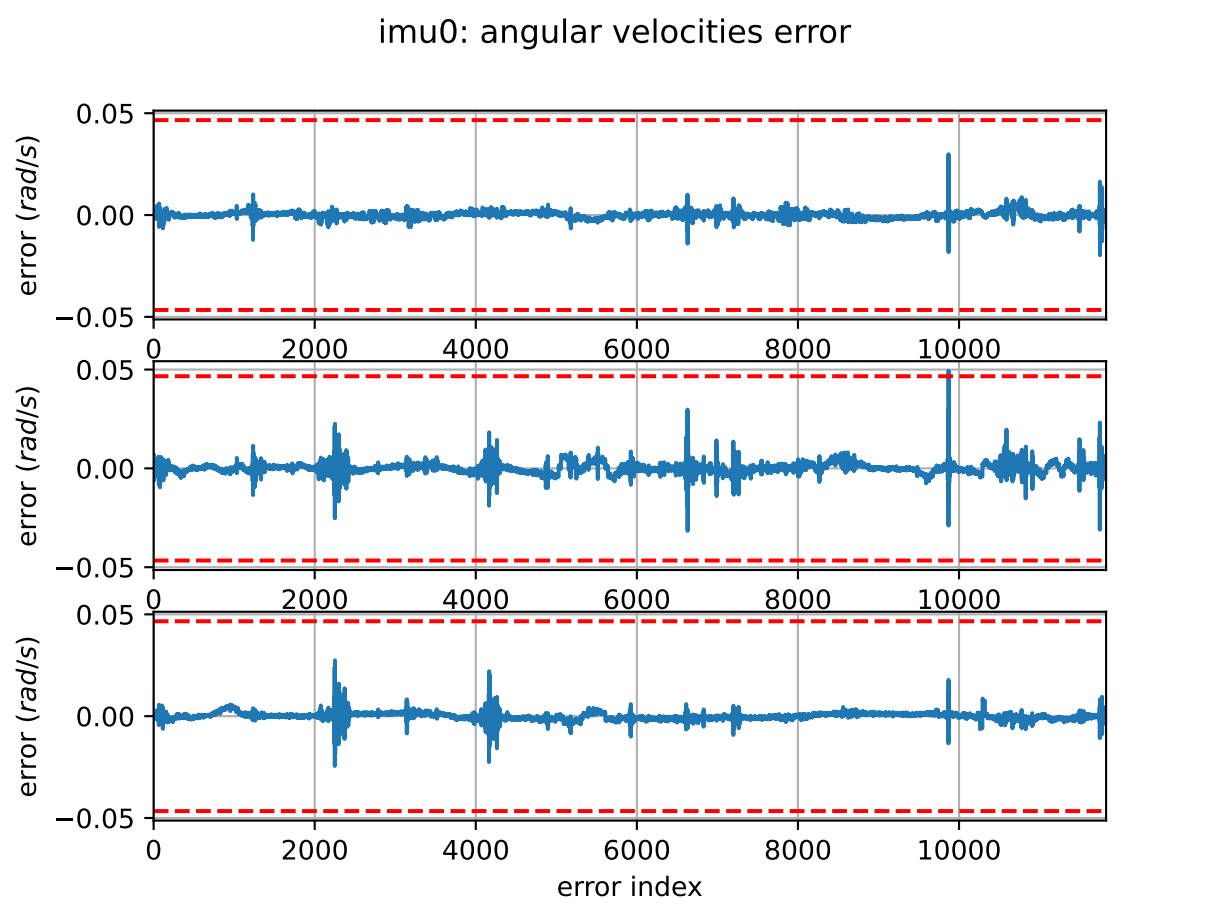

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0001
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
time (s)

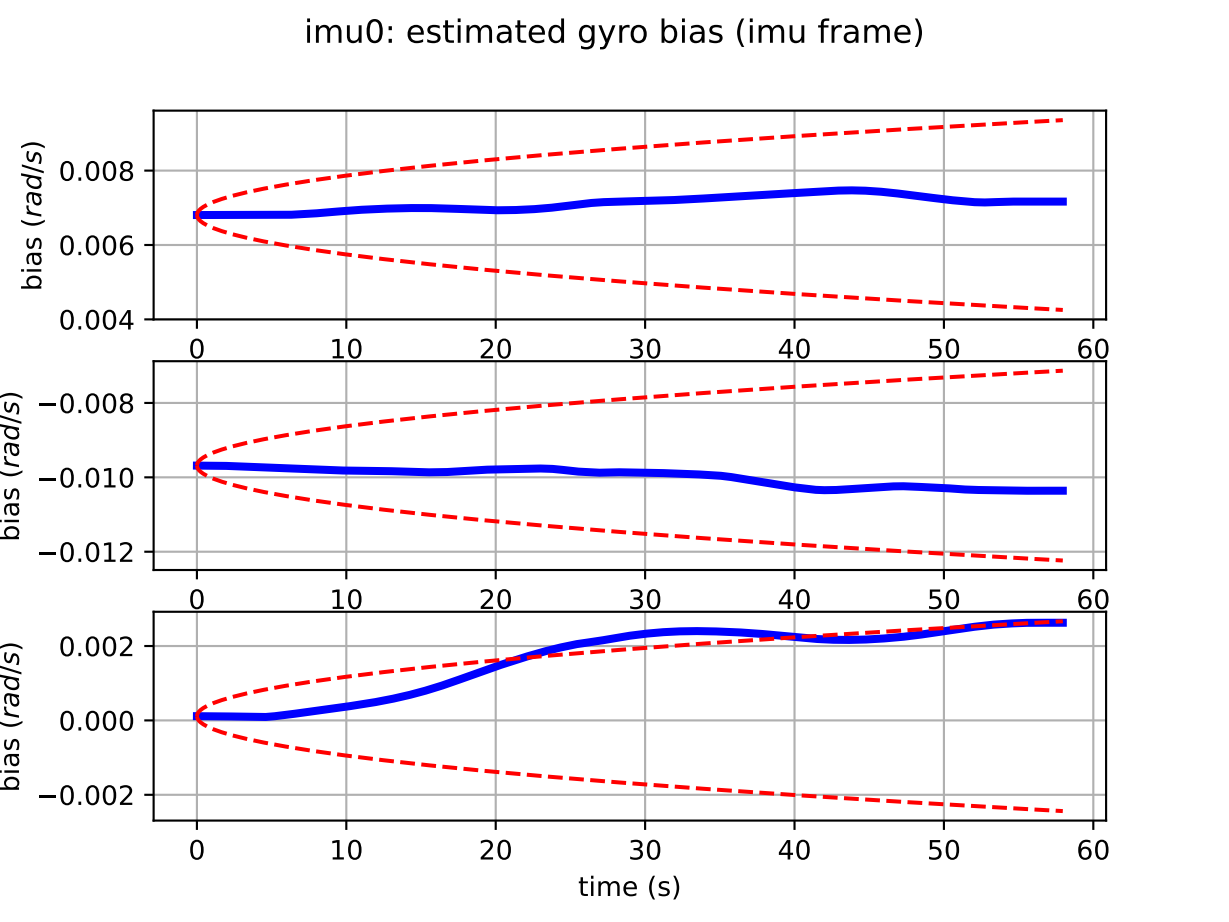

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
time (s)

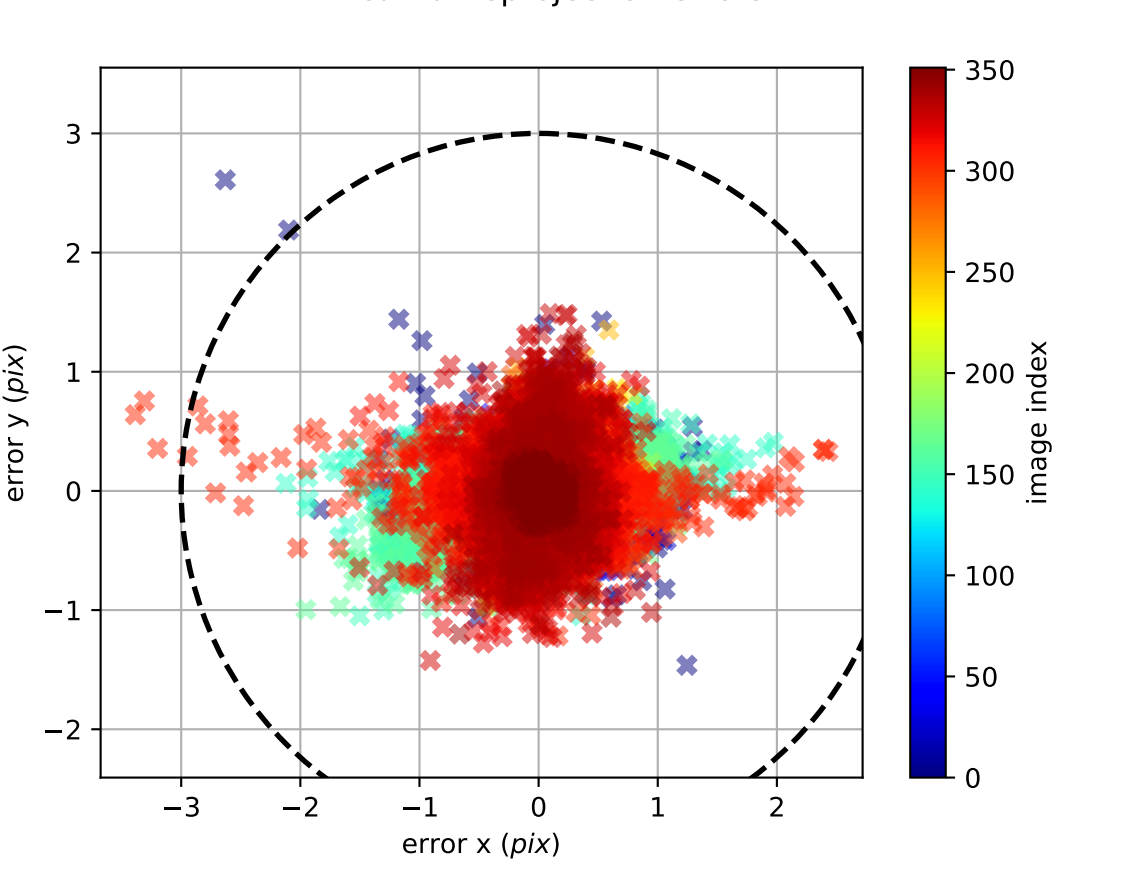

imu0: acceleration error
error (m/s^2)
1
0
−1
0
2000
4000
6000
8000
10000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.25
0.00
−0.25
0
10
20
30
40
50
60
bias (m/s^2)
0.00
−0.25
−0.50
bias (m/s^2)
0.50
0.25
0.00
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
ang. velocity (rad/s)
est. bodyspline
imu0
ang. velocity (rad/s)
est. bodyspline
imu0
time (s)


imu0: angular velocities error
error (rad/s)
0.05
0.00
−0.05
0
2000
4000
6000
8000
10000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.008
0.006
0.004
0
10
20
30
40
50
60
bias (rad/s)
−0.008
−0.010
−0.012
bias (rad/s)
0.002
0.000
−0.002
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−3
−2
−1
0
1
2
3
0
50
100
150
200
250
300
350